

Problem

Does one of the first

$$10^8 + 1$$

Fibonacci numbers terminate with four zeroes?

Equivalently, does there exist an index

$$0 \leq n \leq 10^8$$

such that

$$10^4 \mid F_n?$$

Source: Original Problem by Aryan Goutam

§1 Encoding Fibonacci Numbers by Consecutive Pairs

Consider the Fibonacci sequence

$$F_0 = 0, \quad F_1 = 1, \quad F_{n+2} = F_{n+1} + F_n.$$

To determine whether a Fibonacci number is divisible by 10^4 , it is natural to study the sequence modulo 10^4 .

Instead of examining individual terms, consider the ordered pairs

$$(F_n, F_{n+1}) \pmod{10^4}.$$

Since each coordinate may assume one of 10^4 residue classes, there are at most

$$10^4 \cdot 10^4 = 10^8$$

possible pairs.

Moreover, the Fibonacci recurrence determines the next pair uniquely:

$$(F_n, F_{n+1}) \mapsto (F_{n+1}, F_n + F_{n+1}).$$

Hence the sequence of pairs evolves deterministically within a finite set of 10^8 states.

§2 Why the Sequence Is Purely Periodic

A key observation is that the transformation

$$(a, b) \mapsto (b, a + b)$$

is invertible modulo 10^4 .

Indeed, from a pair (b, c) one recovers the preceding pair as

$$(c - b, b).$$

Therefore every state has a unique predecessor as well as a unique successor.

Consequently, the sequence of pairs modulo 10^4 cannot merely become periodic after a transient phase; it is periodic from the very beginning.

§3 The Pigeonhole Principle at Work

The sequence starts at

$$(F_0, F_1) \equiv (0, 1) \pmod{10^4}.$$

Since there are only 10^8 possible states, among the first

$$10^8 + 1$$

pairs

$$(F_0, F_1), (F_1, F_2), \dots, (F_{10^8}, F_{10^8+1})$$

two must coincide.

Because the sequence is purely periodic, the first repeated state must be the initial state itself. Thus for some integer

$$1 \leq r \leq 10^8$$

we have

$$(F_r, F_{r+1}) \equiv (F_0, F_1) \equiv (0, 1) \pmod{10^4}.$$

In particular,

$$F_r \equiv 0 \pmod{10^4}.$$

Therefore one of the Fibonacci numbers among

$$F_0, F_1, \dots, F_{10^8}$$

is divisible by 10^4 .

Hence one of the first $10^8 + 1$ Fibonacci numbers ends in four zeroes.

Yes, such a Fibonacci number must exist.

§4 A Stronger Result

The argument above gives only an upper bound.

In fact, the sequence of Fibonacci numbers modulo 10^4 has period

$$15000,$$

known as the Pisano period for 10^4 .

A computation shows that

$$10^4 \mid F_{7500}.$$

Thus a Fibonacci number ending in four zeroes appears far earlier than the bound obtained from the pigeonhole principle.

The proof above, however, requires no knowledge of the exact period and relies only on the finiteness and invertibility of the recurrence modulo 10^4 .